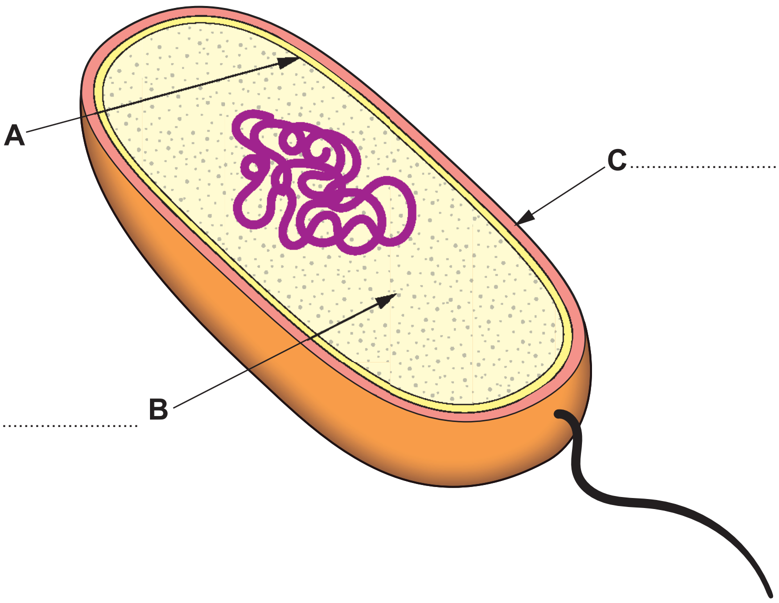
Examiner only

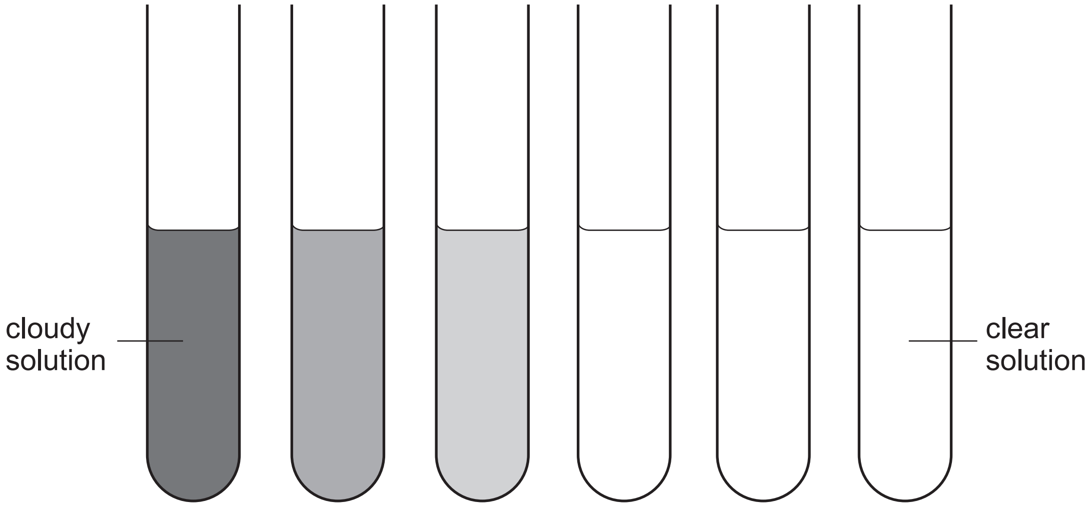
2. The diagram shows a bacterial cell.



(a) Label structures **A**, **B** and **C** on the diagram above. [3]

(b) Carys wanted to investigate the effect of an antibiotic on bacterial growth. She set up the experiment shown below with test tubes containing increasing concentrations of an antibiotic and inoculated each test tube with a fixed volume of nutrient broth containing a known concentration of bacteria.

The tubes were incubated at 25 °C for two days. The cloudier the solution becomes the more bacteria present. A clear sample means there is no bacterial growth.



Tube number	1	2	3	4	5	6
Concentration of antibiotic (μg/cm^3)	0.25	0.50	1.00	2.00	4.00	8.00

Examiner only

(i) Explain the result observed in tubes **4–6**. [1]

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....

(ii) Suggest the minimum concentration of antibiotic that would be effective against the bacterium. [1]

minimum concentration = ... $\mu g/cm^3$

(iii) Suggest how Carys could improve her experiment so that she could obtain a more accurate value for the minimum concentration of antibiotic needed to be effective against the bacterium. [2]

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